C-HI Through-10 Myr Snapshot Series

Date: 2026-05-11

Summary

The reduced-grid particles-enabled C-HI shakedown run has now reached the nominal 10 Myr regular-output block on kunanyi. This report extends the aligned scalar diagnostic lane from the previous 1-5 Myr series through the new regular outputs:

Grid output	Time (Myr)	Particle output
10	5.9997	604
11	6.9990	704
12	7.9993	804
13	8.9996	904
14	9.9999	1004

The same compact lane was run for each snapshot:

- Level 1 injected-energy truth
- Level 2 tracer-defined lobe truth at tracer > 5e-3
- first-pass ellipsoidal cavity protocol
- XRAiSE-corrected 1 keV grid-thermal X-ray scalar smoke
- particle-loss-aware 150 MHz radio/lobe scalar smoke

The particle HDF5 cache was rebuilt first and now spans 0-10.278 Myr, with 1033 particle times and a full HDF5 size of 423 MB on kunanyi.

Kunanyi Jobs

Main continuation:

- job id: 690650.kunanyi-ohpc
- job name: c_hi_to10
- resources: select=2:ncpus=28:mpiprocs=28
- exit status: 0
- walltime: 06:44:13
- output volume after run: 95G
- final regular output: data.0014.flt.h5, 9.999876 Myr
- final restart checkpoint: data.0007.dbl.h5, 10.278362 Myr

Follow-up diagnostics:

- particle HDF5 job: 690680.kunanyi-ohpc, Exit_status=0
- scalar-series job: 690681.kunanyi-ohpc, Exit_status=0
- particle archive SHA256: e674ebe6710910237558e8d3ea6f86716e3ad8b7e73a1b73d4588a555610a03c
- scalar archive SHA256: 731c8e65e0818fa4eb934782a37289a9e9c31e7eb752a5563e9b5c1b51955943

The only diagnostic stderr issue was the known NumbaParallelSafetyWarning from the supplied particle-loss path. This same radio path has already passed a NUMBA_NUM_THREADS=1 versus 4 exact determinism check at the aligned 5 Myr target.

Main Trends

Metric	5 Myr	10 Myr	Change
bipolar injected energy	3.1554e59 erg	6.3114e59 erg	2.00x
Level 2 tracer-lobe energy	1.8459e59 erg	3.4309e59 erg	1.86x
bipolar 4pV cavity proxy	2.5367e59 erg	6.8561e59 erg	2.70x
Level 2 tracer-lobe volume	1.0792e4 kpc^3	3.6574e4 kpc^3	3.39x
tracer z extent	33.04 kpc	47.71 kpc	1.44x
XRAiSE-corrected X-ray peak	2.8783e6 mJy sr^-1	2.8767e6 mJy sr^-1	1.00x
lossy 150 MHz luminosity proxy	6.3450e27 W Hz^-1	7.7444e27 W Hz^-1	1.22x
selected particles	16054	32065	2.00x

At 10 Myr, the compact diagnostics give:

- Level 2 tracer-lobe energy is 0.544 of elapsed bipolar injection
- bipolar 4pV is 1.086 of elapsed bipolar injection
- tracer-defined lobe extent is about +/-47.7 kpc in z
- fitted cavity centres sit at about +/-23.87 kpc
- mean sound age is about 21.69 Myr
- radio lobe balance remains close to unity, with +z / -z = 0.977

Across the 6-10 Myr block, radio lobe balance ranges from 0.946 to 1.047. The radio luminosity proxy is not monotonic, ranging from 6.32e27 to 1.33e28 W Hz^-1, while the central X-ray peak remains essentially flat.

Figures

Products

Compact products are stored under:

```
docs/science/assets/c_hi_to10_snapshot_series_2026-05-11/
docs/science/assets/c_hi_to10_particle_hdf5_2026-05-11/
docs/science/assets/c_hi_to10_combined_series_2026-05-11/
```

Key files:

- c_hi_to10_snapshot_series_metrics.csv
- c_hi_to10_snapshot_series.png
- c_hi_to10_combined_series_metrics.csv
- c_hi_to10_combined_series.png
- compact/snap_0010/
- compact/snap_0011/
- compact/snap_0012/
- compact/snap_0013/
- compact/snap_0014/

Reusable scripts:

- scripts/kunanyi/c_hi_to10_particle_hdf5.pbs
- scripts/kunanyi/c_hi_to10_snapshot_series.pbs
- scripts/science/build_early_snapshot_series.py

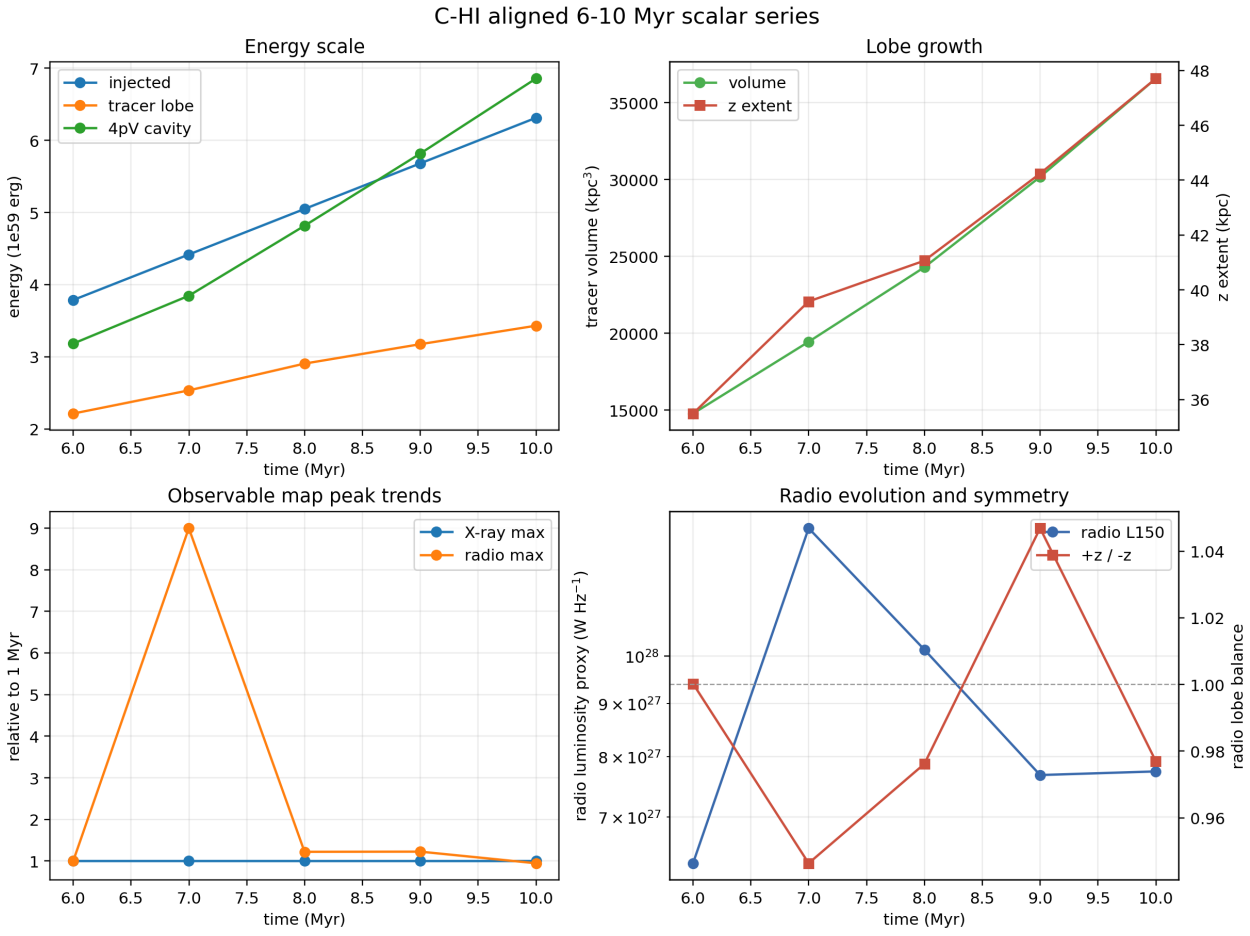


Figure 1: C-HI aligned 6-10 Myr scalar series

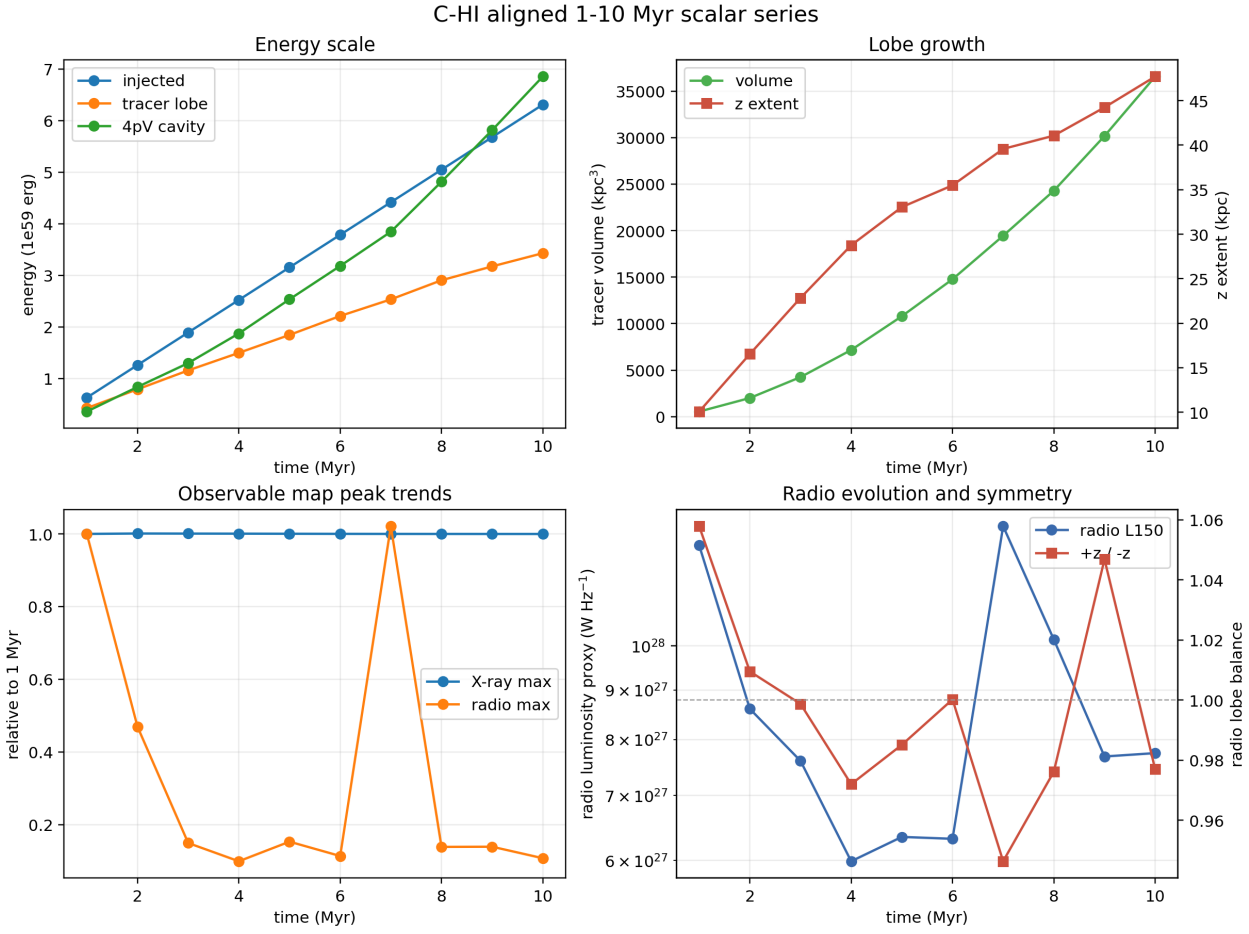


Figure 2: C-HI aligned 1-10 Myr scalar series

Interpretation

This closes the through-10-Myr diagnostic gate for the reduced-grid `C-HI` shakedown. The lobe growth is smooth, the two sides remain well balanced, the jet remains comfortably inside the `+/-250 kpc` axial domain, and the restart catalogue is healthy.

The next compute-facing move can be another bounded continuation from `data.0007.dbl.h5` toward the next regular snapshot block, or a short production-scale allocation test if the goal is to transition from this reduced-grid lane toward the full-grid `80 node` production pattern discussed with Stas and Patrick.